TSANG Cheuk Nam

Hong Kong | rairu0811@gmail.com | rairurairu.github.io | github.com/rairurairu

Summary

Final-year HKUST Industrial Engineering & Engineering Management student (Extended Major in AI) who designs AI systems and the processes around them, directing AI coding agents (Claude Code, Codex) that write about 95% of the code. Internship now; full-time from January 2027.

Education

The Hong Kong University of Science and Technology (HKUST) Hong Kong
BEng in Industrial Engineering & Engineering Management; Extended Major in AI *Sep 2022 – Dec 2026 (expected)*

Honors: Winner, "Reinventing Insurance for an AI-native World" track (sponsored by bolttech), Hong Kong AI Summit: BUILD ACROSS Hackathon, 200+ participants (Oct 2026)

Coursework: Statistical Machine Learning; Generative AI and LLMs for Operations Research; Stochastic Models

Projects

Budget-Based Microinsurance | *Hackathon-winning methodology* · Rust, TypeScript, Python Oct 2026

- Designed an original AI-native insurance methodology: customers pick the price and AI builds the cover, answering the long-standing pain points of traditional insurance.
- Took routine work off insurance agents and claims handlers with AI, from proposal forms to claim assessment.
- Designed where AI stops: it recommends, fixed rules set every price and payout, and only a person can refuse a claim.
- Built every AI component to fall back to a saved answer when a reply fails its checks, and tested each one against deliberately bad replies.
- Built the payout engine to settle automatic payouts for 1 million policies in 1.6 seconds, against a 5-second target.

LLM Brand Recommendation Benchmark | *AI-native market research* · Python, SQLite, LLM APIs Sep 2026

- Turned manual brand research into an AI-run benchmark of which brands AI assistants recommend per category.
- Redesigned the method after the first approach validated only one category; every category but one then passed.
- Found that most failed checks came from incomplete reference lists that marked real brands as wrong, and added only verified brands to the lists.

Using LLMs for Optimization | *IEDA Final Year Project, Experiment & Benchmarking Lead* Completed 2026

- Led the routing experiments showing that an LLM planning for a standard optimizer matched the solution quality of an LLM writing the whole program, at a fraction of the cost.

Self-Hosted Personal AI Agent | *Always-on AI agent and coding-agent team* · Python, Kotlin Aug 2026 – Present

- Wrote the rules his always-on agent and coding agents work under, drawn from their real failures: agents build only what was asked, and only a person can start anything that acts in the real world.

Polymarket Market-Making & Forecasting Models | *AI-agent-driven quantitative research* Aug 2026 – Present

- Built football forecasting models that beat the standard Elo rating on matches they had never seen.
- Replayed historical orders to size risk before quoting: prices hit sell quotes about 4.5× as often as buy quotes.

9UPPER Online | *Multilingual real-time web game* · Next.js, TypeScript, Supabase Mar 2026 – Present

- Built it with AI coding agents and grew it to over 500 visitors a day through SEO, GEO and AEO; ChatGPT and Claude search recommend it.

Skills

AI application design: LLM components, evals and guardrails, structured outputs, model routing, human-in-the-loop

AI-native transformation: process redesign, deciding what AI, rules or people handle, cost and quality measurement

Agentic development: Claude Code, Codex, multi-agent orchestration, harness and context engineering

Software: Python, TypeScript, React, Rust, PostgreSQL, REST APIs, Docker, Linux, Git

Languages: Cantonese (native), English (fluent), Mandarin (fluent)